

# ■ DSA PLACEMENT ROADMAP

3-Month Complete Guide · 250–300 Quality Problems · Placement Ready

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|          |                |           |                |
|----------|----------------|-----------|----------------|
| <b>3</b> | <b>250–300</b> | <b>12</b> | <b>4–6 hrs</b> |
| Months   | Problems       | Weeks     | Daily          |

## ■ OVERALL STRATEGY

|                    |                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------|
| ■■ <b>Language</b> | C++ / Java — pick ONE and stick to it throughout                                                |
| ■ <b>Platform</b>  | <a href="https://leetcode.com">LeetCode (leetcode.com)</a> — your primary daily practice ground |
| ■ <b>Target</b>    | 250–300 quality problems (NOT mindless random grind)                                            |
| ■ <b>Pattern</b>   | Learn → Practice → Revise → Mock (repeat every week)                                            |

## ■ DAILY ROUTINE (4–6 hours)

|                 |                   |             |                    |
|-----------------|-------------------|-------------|--------------------|
| <b>1 hr</b>     | <b>2–3 hr</b>     | <b>1 hr</b> | <b>1 hr</b>        |
| ■ Concept Study | ■ Problem Solving | ■ Revision  | ■ Notes + Mistakes |

## ■ GOLDEN RULES

|                                                         |
|---------------------------------------------------------|
| ■ Don't jump topics — follow the sequence strictly      |
| ■ Re-solve questions — THIS IS THE KEY to mastery       |
| ■ Maintain a mistake notebook for every wrong answer    |
| ■ Focus on PATTERNS, not the number of questions solved |
| ■ After 2 months → START MOCK INTERVIEWS immediately    |

## ■ MONTH 1 — Foundation + Core Logic

### Basics + Arrays

■ Time Complexity & Big-O Notation (VERY IMPORTANT — learn this first) | ■ Arrays — in-place tricks, two pointers, focus on patterns

- #1 Two Sum
- #121 Best Time to Buy & Sell Stock
- #238 Product of Array Except Self
- #15 3Sum
- #11 Container With Most Water
- ...and more on LeetCode
- #53 Maximum Subarray (Kadane's)
- #217 Contains Duplicate
- #88 Merge Sorted Array
- #56 Merge Intervals
- #48 Rotate Image

■ **Goal: Pattern recognition — spot the trick before you start coding**

### Strings + Recursion

■ Strings — manipulation, sliding window, anagrams | ■ Recursion + Backtracking intro — visualise the call tree always

- #20 Valid Parentheses
- #242 Valid Anagram
- #5 Longest Palindromic Substring
- #443 String Compression
- #46 Permutations
- ...and more on LeetCode
- #125 Valid Palindrome
- #49 Group Anagrams
- #14 Longest Common Prefix
- #78 Subsets
- #39 Combination Sum

■ **Goal: Think recursively — draw the call tree for every backtracking problem**

### Sorting + Binary Search

■ Binary Search — VERY IMPORTANT, classic + rotated + answer-space | ■ Merge Sort / Quick Sort — concepts, not just code

- #704 Binary Search
- #81 Search in Rotated Array II
- #74 Search a 2D Matrix
- #875 Koko Eating Bananas
- ...and more on LeetCode
- #33 Search in Rotated Sorted Array
- #34 First and Last Position
- #162 Find Peak Element
- #153 Min in Rotated Array

■ **Goal: Master all binary search templates (classic, rotated, answer space)**

### Linked List

■ Singly & Doubly Linked Lists — basics | ■ Slow / Fast pointer — cycle detection, nth from end

- #206 Reverse Linked List
- #21 Merge Two Sorted Lists
- #2 Add Two Numbers
- #234 Palindrome Linked List
- ...and more on LeetCode
- #141 Linked List Cycle (Floyd's)
- #19 Remove Nth Node From End
- #160 Intersection of Two Lists
- #25 Reverse Nodes in k-Group

■ **Goal: Pointer mastery — always draw node diagrams while solving**

## ■ MONTH 2 — Core DSA (Most Important)

### Stack + Queue

■ Stack fundamentals + Monotonic Stack ■ — very common in interviews | ■ Queue variants — circular queue, deque, two-stack queue

- #20 Valid Parentheses
- #225 Implement Stack using Queues
- #739 Daily Temperatures
- #496 Next Greater Element I
- #622 Design Circular Queue
- ...and more on LeetCode
- #232 Implement Queue using Stacks
- #155 Min Stack
- #150 Evaluate Reverse Polish Notation
- #503 Next Greater Element II
- #84 Largest Rectangle in Histogram

■ **Goal: Instantly recognise when monotonic stack is the right tool**

### Trees ■ (SUPER IMPORTANT)

■ DFS & BFS traversals — know all 4 cold (pre, in, post, level) | ■ Binary Tree problems — recursion is your best friend here

- #144 Preorder Traversal
- #145 Postorder Traversal
- #104 Maximum Depth
- #101 Symmetric Tree
- #543 Diameter of Binary Tree
- ...and more on LeetCode
- #94 Inorder Traversal
- #102 Level Order Traversal (BFS)
- #100 Same Tree
- #110 Balanced Binary Tree
- #124 Max Path Sum

■ **Goal: Most tree problems = a traversal variant — build that intuition**

### BST + Heaps

■ BST — in-order = sorted, search/insert/delete patterns | ■ Priority Queue / Heap — know the API, when it beats sorting

- #700 Search in BST
- #230 Kth Smallest in BST
- #215 Kth Largest Element
- #23 Merge K Sorted Lists
- ...and more on LeetCode
- #98 Validate BST
- #235 LCA of BST
- #347 Top K Frequent Elements
- #295 Find Median from Data Stream

■ **Goal: Know exactly when a heap beats sorting (real-time k-th problems)**

### Hashing + Sliding Window ■

■ Hash Map / Hash Set —  $O(1)$  lookup, frequency maps, prefix sums | ■ Sliding Window — MOST ASKED pattern in product company interviews

- #128 Longest Consecutive Sequence
- #49 Group Anagrams
- #3 Longest Substring Without Repeating
- #424 Longest Repeating Character Replacement
- ...and more on LeetCode
- #560 Subarray Sum Equals K
- #1 Two Sum (hash variant)
- #76 Minimum Window Substring
- #567 Permutation in String

■ **Goal: Optimise  $O(n^2)$  brute force  $\rightarrow O(n)$  using hash map or sliding window**

## ■ MONTH 3 — Advanced + Interview Prep

### Graphs (VERY IMPORTANT)

■ BFS and DFS on graphs — adjacency list & matrix both | ■ Topological Sort + Union-Find (Disjoint Set Union)

- #200 Number of Islands
- #207 Course Schedule
- #785 Is Graph Bipartite?
- #323 Number of Connected Components
- ...and more on LeetCode
- #133 Clone Graph
- #210 Course Schedule II
- #994 Rotting Oranges
- #417 Pacific Atlantic Water Flow

■ **Goal: Model any network / dependency problem as a graph first**

### Dynamic Programming ■

■ 1D DP → 2D DP → space optimisation — build logic step by step | ■ Identify overlapping subproblems — do NOT memorise formulas

- #70 Climbing Stairs
- #322 Coin Change
- #1143 Longest Common Subsequence
- #416 Partition Equal Subset Sum
- #718 Max Length of Repeated Subarray
- ...and more on LeetCode
- #198 House Robber
- #300 Longest Increasing Subsequence
- #62 Unique Paths
- #5 Longest Palindromic Substring
- #53 Maximum Subarray

■ **Goal: Pattern recognition — NOT memorisation of solutions**

### Greedy + Bit Manipulation + Revision

■ Greedy — prove locally optimal = globally optimal | ■ Bit Manipulation + Re-solve ALL starred problems from Weeks 1–10

- #55 Jump Game
- #134 Gas Station
- #136 Single Number (XOR)
- #191 Number of 1 Bits
- #435 Non-overlapping Intervals
- ...and more on LeetCode
- #45 Jump Game II
- #253 Meeting Rooms II
- #190 Reverse Bits
- #268 Missing Number
- #621 Task Scheduler

■ **Goal: Connect patterns across topics, spot the overlap between them**

### Mock + Interview Mode ■

■ Daily timed LeetCode contests — 35 min per problem max | ■ Peer mocks — code AND communicate simultaneously, every single day

- Blind 75 — complete any remaining problems
- 1 full 60-minute mock interview every day
- Revise your mistake notebook thoroughly
- ...and more on LeetCode
- 2–3 mixed difficulty problems per day (timed)
- Review every HARD problem attempted
- Company-specific LeetCode sets (Amazon, Google, etc.)

■ **Goal: Simulate real interview pressure — performance under stress**

## ■ TARGET BREAKDOWN

| Level    | Count     | Purpose             |
|----------|-----------|---------------------|
| ■ Easy   | 80 – 100  | Foundation Building |
| ■ Medium | 120 – 150 | Core Interview Prep |
| ■ Hard   | 30 – 40   | Bonus / FAANG Prep  |
| ■ TOTAL  | 250 – 300 | Placement Ready ■   |

## ■ RESOURCES — Everything You Need to Crack Placements

### ■ PRACTICE PLATFORMS

|                |                                                       |                                                                                             |
|----------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------|
| ■ LeetCode     | <a href="https://leetcode.com">leetcode.com</a>       | Primary platform — solve all 75+ questions here. Use the Discuss section for top solutions. |
| ■ Syntax Error | <a href="https://syntaxerrrr.com">syntaxerrrr.com</a> | Our website — notes, one-shot videos, placement series and all DSA resources.               |

### ■■ YOUTUBE RESOURCES

|                     |                                                                                                                                                                                                                                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ■ PRIMARY RESOURCE  | <b>Striver — A to Z DSA Sheet Playlist</b> by takeUforward<br><a href="https://youtube.com/playlist?list=PLgUwDviBlf0oF6...">youtube.com/playlist?list=PLgUwDviBlf0oF6...</a><br>The gold standard for DSA prep. Arrays, Trees, Graphs, DP — everything. 450+ videos, structured from basics to advanced. Follow this religiously. |
| ■ DP + RECURSION    | <b>DP &amp; Recursion Masterclass</b> by Aditya Verma<br><a href="https://youtube.com/@TheAdityaVerma">youtube.com/@TheAdityaVerma</a><br>The best playlist to build recursion intuition and DP from scratch. Start here for Week 10 DP topics — extremely clear explanations, must watch.                                         |
| ■ ONE-SHOT REVISION | <b>Syntax Error — One Shot Placement Series</b> by Syntax Error<br><a href="https://youtube.com/@SYNTAXERROR-code">youtube.com/@SYNTAXERROR-code</a><br>Our own placement series — all DSA topics covered in one-shot format. Perfect for quick revision before exams and final interviews.                                        |

### ■ PRO TIPS FROM SYNTAX ERROR

|   |                                                                                  |
|---|----------------------------------------------------------------------------------|
| ■ | Striver's A to Z sheet is your BIBLE — follow it topic by topic without skipping |
| ■ | Use Aditya Verma for DP & Recursion — don't skip, watch every single video       |
| ■ | Our one-shot videos on syntaxerrrr.com are perfect for last-week quick revision  |
| ■ | Discuss section on LeetCode > reading editorial — always check top solutions     |
| ■ | Consistency beats intensity — 2 hrs daily for 3 months > 10 hrs for 2 weeks      |

|                     |                                                                      |                                                                   |
|---------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>SYNTAX ERROR</b> | Our Website<br><a href="https://syntaxerrrr.com">syntaxerrrr.com</a> | Our Instagram<br><a href="https://code.abhii07">@code.abhii07</a> |
|---------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|

*"The expert in anything was once a beginner. One consistent month beats three inconsistent months.  
Show up EVERY DAY. ■"*